COURSEWORK ASSIGNMENT SUBMISSION

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Subject: Software Engineering & Project Management **Day:** Day 23 — COCOMO Effort Estimation & Project Risk Management

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Section 11: Classroom Activity — Estimate and Manage Risk

Algorithmic software cost and effort estimation for the College Event Management System using the Basic COCOMO Model.

1. Basic COCOMO Formulae & Project Mode

- **Project Mode:** Organic Mode (Small, experienced team working in a familiar in-house campus software environment with relaxed specifications).
- **Standard Organic Coefficients:** $a = 2.4, b = 1.05, c = 2.5, d = 0.38$
- **Size Estimation:** Estimated lines of source code = 4,000 Source Lines of Code (4.0 KLOC).

2. Effort, Duration & Staffing Calculations

Effort (Person-Months): $E = a \times (KLOC)^b = 2.4 \times (4.0)^{1.05} = 2.4 \times 4.287 = 10.29$ Person-Months.
Development Time (Months): $D = c \times (E)^d = 2.5 \times (10.29)^{0.38} = 2.5 \times 2.434 = 6.09$ Months.
Average Staffing: $S = \frac{E}{D} = \frac{10.29}{6.09} \approx 1.69 \approx 2$ Full-Time Engineers.
Cost Calculation: Assuming an institutional developer cost rate of Rs. 60,000 per person-month: Total Labor Cost = $10.29 \times 60,000 = \text{Rs. } 6,17,400$.

3. Project Risk Register (Top 5 Risks)

Risk ID	Cause -> Event -> Impact	Prob.	Impact	Risk Exposure	Owner & Mitigation
R-01	Unfamiliarity with WebSocket -> Integration delay -> Schedule slippage	Medium	High	High	Lead Dev: Conduct 2-day proof-of-concept spike in Sprint 1.
R-02	Sudden requirement change -> Unapproved scope creep -> Budget overrun	High	Medium	High	Project Lead: Enforce strict CCB change management protocol.
R-03	Database un-indexed joins ->	Medium	High	High	DBA: Mandate query explain

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	Server crash under surge -> System outage				plans and stress tests in staging.
R-04	Key developer leaves -> Knowledge loss -> Delivery stalls	Low	High	Medium	Manager: Mandate daily Git commits and modular documentation.
R-05	Campus Wi-Fi downtime -> Gate check-in failure -> Long queues	High	Low	Medium	QA Lead: Implement offline localStorage QR cache on scanners.

Section 14: Homework — COCOMO Calculations & Brooks' Law Analysis

Mathematical effort calculation for a 5 KLOC system and theoretical examination of human factors in project scheduling.

1. 5 KLOC Organic COCOMO Working

Given: $KLOC = 5.0$, Organic mode ($a = 2.4, b = 1.05, c = 2.5, d = 0.38$).

1. Effort Calculation: $E = 2.4 \times (5.0)^{1.05} = 2.4 \times 5.429 = 13.03$ Person-Months.
2. Development Schedule: $D = 2.5 \times (13.03)^{0.38} = 2.5 \times 2.651 = 6.63$ Months.
3. Recommended Staff Size: $Staff = \frac{E}{D} = \frac{13.03}{6.63} \approx 1.96 \approx 2$ Engineers.

2. Examination of Brooks' Law

- **Brooks' Law:** "Adding manpower to a late software project makes it later."
- **Engineering Explanation:** Software development is fundamentally a complex communication and design exercise, not a divisible manual task like harvesting crops. When new engineers join a late project: (1) Existing productive engineers must divert time from development to onboard and train the new personnel; (2) Communication channels scale quadratically according to $n(n - 1) / 2$, dramatically increasing meeting overhead and coordination friction; (3) Complex task partitioning introduces architectural misunderstanding and defect divergence.